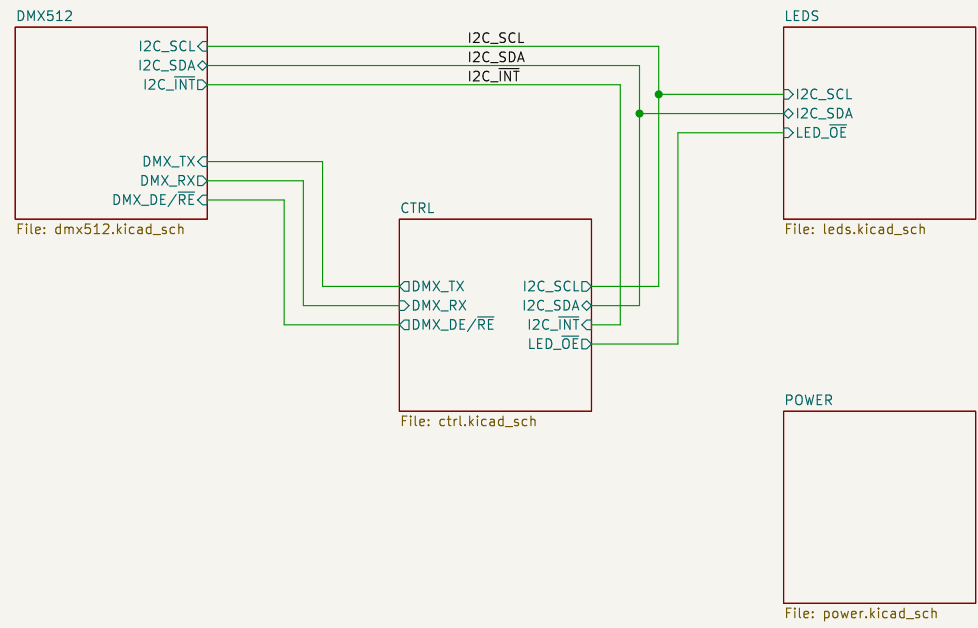


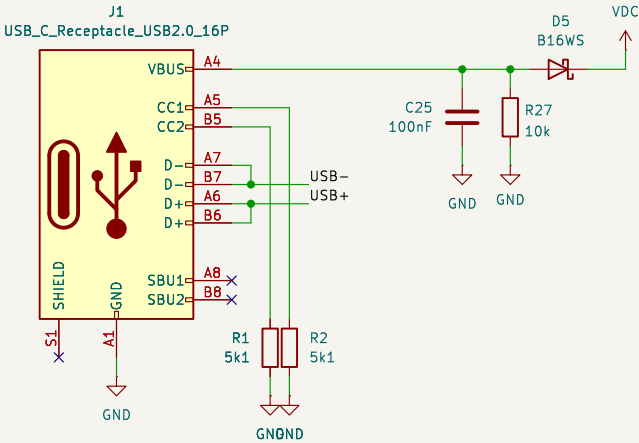
LED Beacon



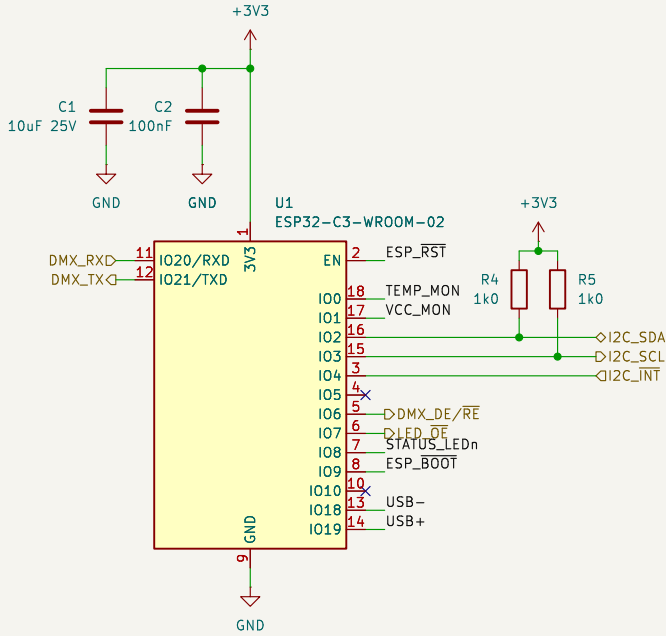
- H1 MountingHole
- H2 MountingHole
- H3 MountingHole
- H4 MountingHole

Wires & Bits		
Sheet: /		
File: emf_beacon.kicad_sch		
Title: LED Beacon		
Size: A4	Date: 2026-06-05	Rev: 1
KiCad E.D.A. 9.0.9		Id: 1/5

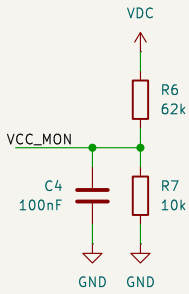
USB Programming



Controller

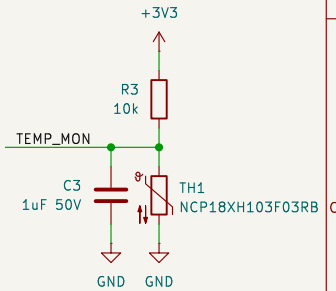


VCC Monitoring

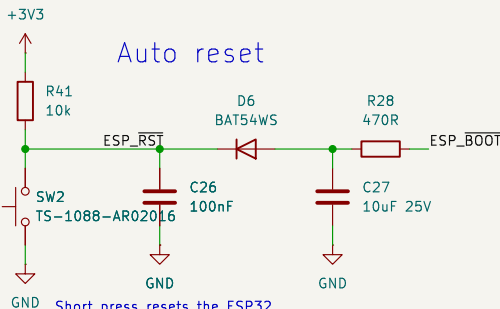


Voltage Monitor GND GND
Vmax is 17.5V
 $V_{adc} \times ((43k\Omega + 10k\Omega) \div 10k\Omega) = VCC$
 $V_{adc} \times 5.3 = VCC$
 $1lsb = 4,271\mu V$

Temperature Monitoring

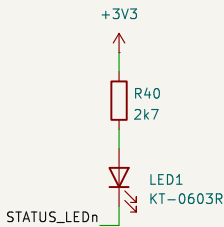


Auto reset



Short press resets the ESP32
Long press resets the ESP32
and puts it in bootloader mode

Status LED



IO8 needs to be held high to boot
so LED & resistor are high side.
Set STATUS_LEDn to 0V to light LED.

Wires & Bits

Sheet: /CTRL/
File: ctrl.kicad_sch

Title: Controller

Size: A4	Date: 2026-06-05
KiCad E.D.A. 9.0.9	

Rev: 1
Id: 2/5

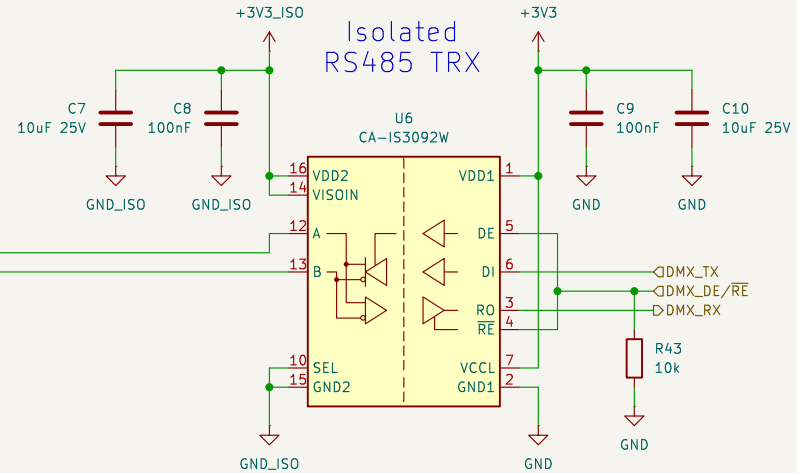
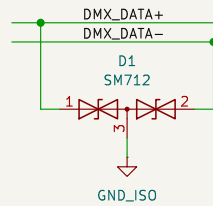
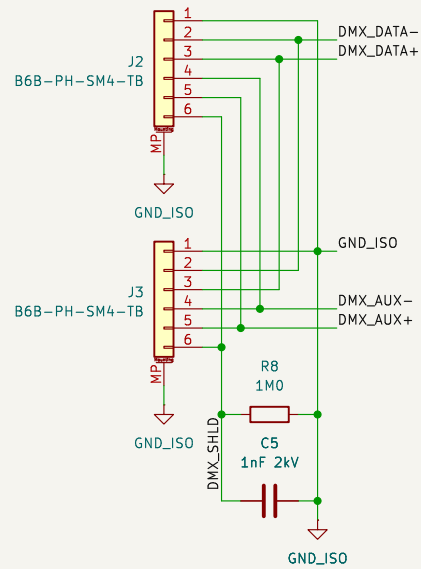
DMX512

Failsafe
Bias
+3V3_ISO

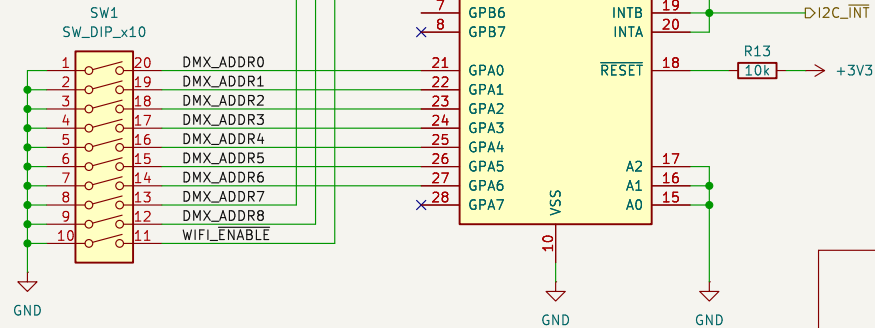
ESD
Protection

Isolated
RS485 TRX

To XLRs



CA-IS3092W has pin 14 as NC
but to maintain compatibility
with CA-IS3092VW it's
connected to 3V3_ISO



I²C Addr: 0x20
MCP23017 has a fault with GPA7 & GPB7.
The can corrupt an I²C transaction if
they change state during it.

Wires & Bits

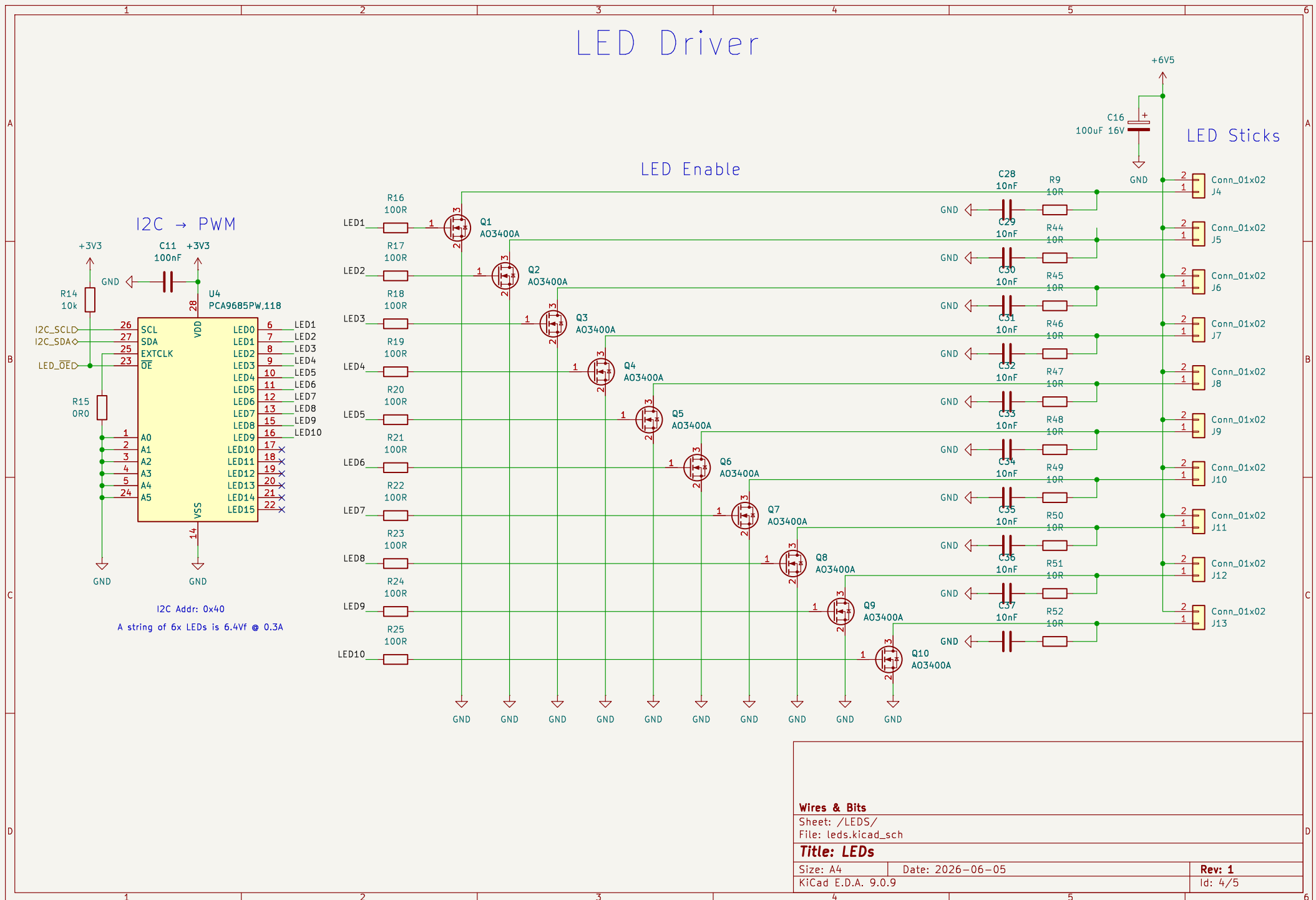
Sheet: /DMX512/
File: dmx512.kicad_sch

Title: DMX512

Size: A4 Date: 2026-06-05
KiCad E.D.A. 9.0.9

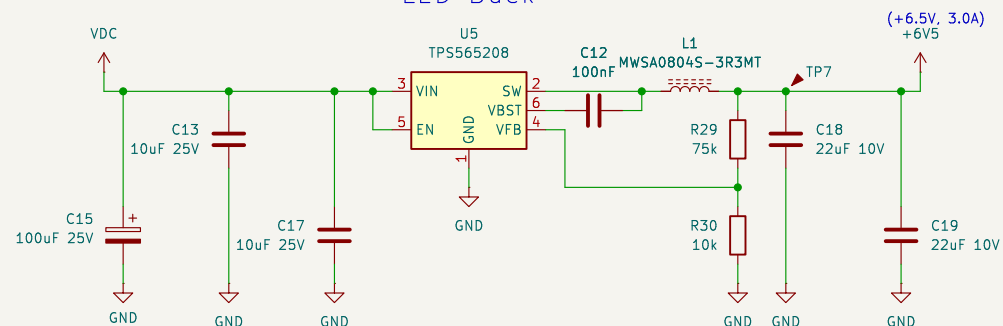
Rev: 1
Id: 3/5

LED Driver

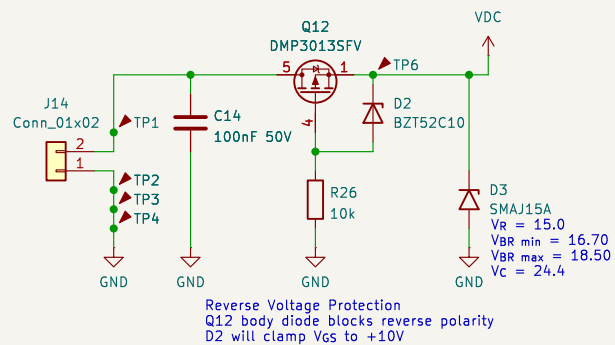
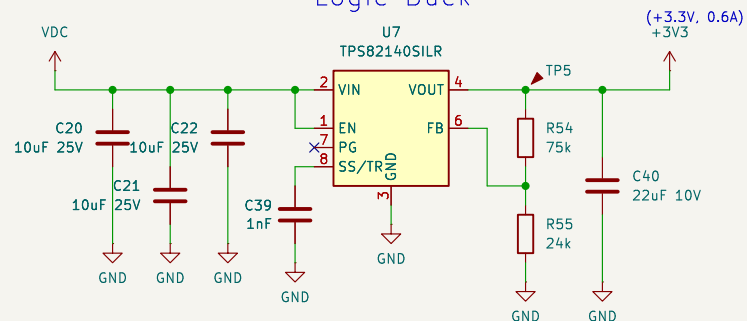


Power Supplies

LED Buck



Logic Buck



EMC ferrites /CMC?

Wires & Bits

Sheet: /POWER/
File: power.kicad_sch

Title: Power

Size: A4
Date: 2026-06-05
KiCad E.D.A. 9.0.9

Rev: 1
Id: 5/5